

Effects of Barley Grain Processing on Extent of Digestion and Milk Production of Lactating Cows¹

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ABSTRACT

Effects of barley processing on site and extent of digestion and milk production in dairy cows were evaluated in a 4 × 4 Latin square design with four lactating cows with ruminal and duodenal cannulas. Barley grain was steam-rolled to four thicknesses: coarse, medium, medium-flat, and flat. The processing index (PI), measured as volume weight of barley after processing expressed as a percentage of its volume weight before processing, was 81.0, 72.5, 64.0, and 55.5% for the four treatments, respectively. Diets consisted of 53% concentrate (dry matter basis) containing one of the four processed barleys. Cows were offered *ad libitum* access to a total mixed ration three times daily. Dry matter intake was quadratically increased with decreasing PI, with maximum intake for cows fed medium-flat barley. Although ruminal digestibilities of organic matter, starch, and crude protein were not affected by grain processing, intestinal and total tract digestibilities were linearly increased as PI of barley was reduced. Milk yield was quadratically increased (25.6, 28.1, 30.8, and 29.0 kg/d) with decreasing PI, and maximum milk yield was for cows fed medium-flat barley. Milk fat and lactose contents were similar, but milk protein content was increased with decreasing PI. These results indicate that the optimal extent of barley processing for dairy cows fed diets supplying adequate fiber was medium-flat, corresponding to a processing index of about 64%. Coarsely or flatly rolled barley is not recommended, because extensive processing did not further improve intake of digestible nutrients, and coarsely processed barley resulted in the lowest intake of digestible organic matter; hence, lowest milk production. Processing index is a reliable and practical method to quantitatively measure extent of steam rolling.

(Key words: barley, grain processing, digestibility, dairy cows)

Abbreviation key: ERD = effective ruminal degradability, PI = processing index measured as volume weight of barley after processing, expressed as a percentage of its volume weight before processing.

INTRODUCTION

Barley is a dominant feed grain in western North America and Europe. Barley processing is imperative to maximize its utilization by feedlot and dairy cattle. Whole grain with an intact pericarp is largely or entirely resistant to digestion by ruminants because whole kernels are resistant to bacterial attachment (5) in the rumen. In addition to pericarp, barley grain is surrounded by a fibrous hull, which is of low digestibility. While unprocessed corn can be effectively fed to ruminants because the pericarp of the kernel is extensively damaged by chewing (6), barley kernels are not severely damaged by chewing. Consequently, considerable whole barley kernels are excreted in feces if whole barley kernels are fed.

In the rumen, barley starch is more readily degradable than corn starch once the pericarp is cracked by processing, because the protein matrix in barley is readily solubilized and penetrated by proteolytic bacteria. However, rapid and extensive ruminal carbohydrate fermentation increases the incidence of bloat, acidosis, laminitis, liver abscesses, and feed intake problems related to digestive upsets.

Physical processing techniques such as grinding or rolling increase digestibility of grain (13, 36), and the advantages and disadvantages of these processing methods are well documented (17, 37). However, there is very little information concerning the optimal degree of processing barley fed to dairy cows. Yang et al. (36) observed that degree of rolling hull-less barley affected milk production and ruminal digestibility. Hironaka et al. (14) concluded that medium steam-rolled barley resulted in better performance than thin, coarse, or whole barley fed to steers. Commercially, barley is often coarsely rolled to ensure a reduced rate of ruminal digestion; however, it is clear that this practice needs to be re-evaluated because the performance of cattle may be dependent on the size of the kernels produced in the

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rolling process. The objectives of this study were to evaluate the effects of the extent of rolling barley on intake, chewing activity, rumen fermentation pattern, site and extent of digestion, and performance of dairy cows.

MATERIALS AND METHODS

Cows and Diets

Four lactating Holstein cows that were surgically fitted with ruminal and duodenal cannulas were used. The ruminal cannulas measured 10 cm in diameter and were constructed of soft plastic (Bar Diamond, Parma, ID). Duodenal cannulas were T-shaped and were placed proximal to the common bile and pancreatic duct, approximately 10 cm distal to the pylorus. At the start of the experiment, the cows averaged 692 ± 51 kg of BW and 72 ± 30 DIM and were housed in individual tie stalls and milked twice daily in their stalls at 0700 and 1700 h. Cows were fed ad libitum a total mixed diet three times daily at 0630, 1500, and 1800 h. Cows were weighed at approximately 0830 h at the beginning and end of each period, and these weights were used to calculate mean BW of cows for each experimental period. Cows were cared for according to the Canadian Council on Animal Care Guidelines (Ottawa, ON, Canada).

The experimental design was a 4×4 Latin square with four 21-d periods. Each period consisted of 10-d of adaptation to diets. A relatively short adaptation period was considered adequate because the diets consisted of the same forage to concentrate ratio. Experimental measurements included milk production and composition, feed intake, chewing activity, ruminal fermentation characteristics, rate of passage of particulate and liquid phases out of the rumen, microbial protein synthesis, site and extent of digestion, and in sacco ruminal digestibility. Each period, cows were fed one of four diets. The four diets consisted of approximately 53% concentrate, 31% barley silage, 8% alfalfa haylage, and 8% chopped alfalfa hay (Table 1, DM basis), and the four diets differed by degree of processing of barley grain (Table 2). The diets were formulated with the Cornell Net Carbohydrate and Protein System (Version 3, 12) to supply adequate metabolizable energy and protein for a 600-kg cow producing 40 kg/d of milk with 3.5% fat.

Barley grain from one source was used throughout the experiment. The barley was first screened to remove chaff and small kernels and then steam-rolled to a coarse, medium, medium-flat, or flat thickness. Degree of processing was quantified using a processing index (PI) measured as the volume weight of the barley after processing, expressed as a percentage of its volume

Table 1. Ingredients and chemical composition of the total mixed diet.

Item	% of DM
Ingredient	
Barley silage ¹	30.5
Alfalfa haylage ¹	8.1
Alfalfa chopped hay ¹	8.1
Barley, steam-rolled	42.5
Corn gluten meal	2.44
Blood meal	0.81
Soybean meal	2.64
Canola meal	0.18
Urea	0.33
Molasses, beet	0.20
Calcium carbonate	0.81
Dicalcium phosphorus	0.41
Monophosphorus	0.08
Vitamin-mineral mix ²	2.24
Chemical	
DM	55.4
OM	91.5
CP	16.7
NDF	35.6
NDF from forages	22.1
Starch	31.7
NE _L , Mcal/kg ³	1.68

¹Chemical composition of alfalfa haylage, barley silage and chopped alfalfa hay (DM basis) was 88.4, 91.2 and 91.4% OM; 52.9, 45.0 and 50.2% NDF; 15.1, 13.9 and 15.9% CP, respectively. Starch content of barley silage was 21.7%.

²Contained 51.97% NaCl, 35.98% Dynamate (Pitman Moore, Inc., Mundelein, IL); 18% K, 11% Mg, 22% S, 1000 mg/kg Fe, 2% ZnSO₄·H₂O, 2.4% MnSO₄·4H₂O, 0.01% CoSO₄·6H₂O, 0.009% Na₂SeO₃, 0.012% ethylenediamine dihydroiodide, 0.8% CuSO₄·5H₂O, 680,000 IU of vitamin A/kg, 160,000 IU of vitamin D/kg, and 2,000 IU of vitamin E/kg.

³Estimated from (23).

weight before processing. Barley was steamed by high pressure for about 5 min before passing through a roller mill (30.5-cm diameter, 20 grooves per 2.54 cm). We used a gauge on the roller and visual inspection to compare thickness of rolled grain to maintain uniform thickness from batch to batch. The steamed barley was dried immediately in a horizontal drier after passing through the roller mill. Kernel thickness and width were measured with a micrometer caliper on 10 kernels for each processed barley. Particle size of processed barley was determined by dry sieving with an oscillating sieve shaker (Analysette 3; Fritsch, Oberstein, Germany) equipped with sieves (W. S. Tyler, Inc., Mentor, OH) arranged in descending mesh size (3.35, 2.36, 1.18, 0.60 mm, and pan). Geometric mean length and standard deviation of rolled barley kernels were determined according to the procedure of the American Society of Agricultural Engineers (1).

Feed offered andorts were measured and recorded daily during the last 10 d of the period to calculate feed intake. Feed samples were collected once weekly, and orts were collected twice weekly for DM determination.

Samples were ground through a 1-mm diameter screen (standard model 4, Arthur Thomas Co., Philadelphia, PA) and composited by period for analysis of OM, NDF, ADF, starch, and CP. Milk production was recorded daily, a.m. and p.m., during the last 10 d of the period and was sampled on 4 consecutive d each week. Milk samples were preserved with potassium dichromate, stored at 4°C, and sent to the Central Alberta Milk Testing Laboratory (Edmonton, AB, Canada) for milk fat, CP, and lactose determination (2) by using an infrared analyzer (Milk-O-Scan 605, Foss Electric, Denmark).

Chewing Activities

Chewing activities of the four cows were monitored visually every 5 min for a 24-h period immediately after the adaptation phase. The assumption was made that the particular chewing activity persisted for the entire 5-min period between each visual observation. Chewing activities were expressed as total minutes for the 24-h period or on the basis of DMI and NDF intake by dividing minutes of eating or ruminating by intake.

Ruminal Fermentation

Rumen pH was measured by an industrial electrode (model PHCN-37; Omega Engineering, Stanford, CT) that was manually introduced into the ventral sac and ruminal mat every 3 h for a 24-h period. Hours during which pH was below 6.2 or 5.8 were calculated assum-

ing that change of pH between two measuring times was linear. The lowest pH for each cow over the entire period was recorded. Ruminal fluid was collected for 1 d at 0900, 1200, and 1600 h from multiple sites in the rumen and pooled by time for each cow. Samples were immediately squeezed through four layers of cheesecloth with a mesh size of 250 μ m. Five milliliters of filtrate was preserved by adding 1 ml of 25% H₃PO₃ to determine VFA, and 9 ml of filtrate was preserved by adding 1 ml of 1% H₂SO₄ to determine NH₃ N. The samples were subsequently stored frozen at -20°C until analyses.

Rate of Passage

Ruminal passage kinetics were measured with Cr-mordanted NDF and Co-EDTA as forage and liquid markers, respectively. Fiber was prepared by repeatedly soaking a mixture of alfalfa hay and barley silage in dilute detergent and rinsing until the NDF content of the material exceeded 80%. Fiber was then dried at 55°C. Methods used to mordant Cr to plant cell walls and to prepare Co-EDTA were those of Udén et al. (31). An amount of 250 g of Cr-mordanted NDF and 300 ml of solution containing 15 g of Co-EDTA were introduced in the rumen via the ruminal cannulas. A representative sample of rumen contents was collected from each cow at 0, 1, 2, 3, 4, 6, 8, 10, 12, 15, 19, 24, 32, and 48 h after dosing with the markers. Samples were squeezed through four layers of cheesecloth. The particles were dried at 55°C, ground through a 1-mm diameter screen

Table 2. Characteristics of barley grain.

Item	Degree of barley processing					SE	P<
	Whole	Coarse	Medium	Medium-flat	Flat		
DM, %	88.1 ^a	83.9 ^c	84.2 ^b	84.2 ^b	84.3 ^b	0.05	0.01
Nutrient content, % of DM							
OM	97.4 ^b	97.7 ^{ab}	97.8 ^a	97.6 ^{ab}	97.6 ^{ab}	0.1	0.09
NDF	26.5 ^a	26.7 ^a	26.7 ^a	25.9 ^b	25.4 ^{ab}	0.5	0.09
ADF	7.1 ^b	7.5 ^{ab}	7.2 ^{ab}	7.9 ^a	8.0 ^a	0.3	0.10
Starch	59.1	59.3	58.4	57.6	59.5	2.2	0.75
CP	13.5	13.5	13.4	13.6	13.3	0.3	0.89
Volume weight, kg/hL	57.2 ^a	46.3 ^b	41.4 ^c	36.6 ^d	31.7 ^e	0.3	0.01
Processing index, ¹ %	100.0 ^a	81.0 ^b	72.5 ^c	64.0 ^d	55.5 ^e	0.4	0.01
Kernel thickness, mm	2.16 ^a	1.92 ^b	1.77 ^c	1.56 ^d	1.47 ^d	0.05	0.03
Kernel width, mm	2.99 ^d	4.13 ^c	4.66 ^b	4.99 ^b	5.84 ^a	0.14	0.02
DM retention on sieve, %							
3.35 mm		54.89 ^c	73.41 ^b	84.79 ^a	87.02 ^a	1.26	0.01
2.36 mm		43.58 ^a	25.22 ^b	13.63 ^c	9.69 ^d	0.95	0.05
1.18 mm		1.46 ^b	1.28 ^b	1.29 ^b	2.90 ^a	0.27	0.02
0.60 mm		0.16 ^b	0.30 ^{ab}	0.30 ^{ab}	0.50 ^a	0.09	0.05
Geometric mean length, mm		6.12 ^a	7.11 ^b	7.76 ^c	7.83 ^c	0.10	0.01
Standard deviation, mm		1.51 ^a	1.43 ^{ab}	1.38 ^b	1.39 ^b	0.03	0.05

^{a,b,c,d,e}Means within the same row not followed by the same letter differ ($P < 0.05$).

¹Processing index is the volume weight of the barley after processing, expressed as a percentage of its volume weight before processing.

(standard model 4), and stored for Cr analyses, while filtrate was centrifuged at $26,000 \times g$ for 20 min and the supernatant was retained for Co determination. Rate of passage out of the rumen for liquid or forage particles were estimated using the formula:

$$C = C_0 e^{-kt}$$

where

- C = marker concentration in the rumen at sampling time t (mg/kg)
- C_0 = marker concentration in the rumen at 0 time (mg/kg)
- k = rate of passage (%/h), and
- t = sampling time after marker dosing (h).

Duodenal Flow and Apparent Digestion

Duodenal flow and apparent digestion of nutrients in the total tract were determined with YbCl_3 (Rhône-Poulenc Inc., Shelton, CT). Ammonia ^{15}N [$(^{15}\text{NH}_4)_2\text{SO}_4$, 10.6% atom % ^{15}N ; Isotec, Miamisburg, OH] was used as a ruminal microbial marker. Marker solution was continuously infused into the rumen via ruminal cannulas with an automatic pump during the last 2 wk of the period. Daily amounts infused were 4 g of Yb and 180 mg of ^{15}N dissolved in 650 ml of water for each cow. Ruminal, duodenal, and fecal samples were collected four times daily every 6 h moving ahead 2 h each day for the last 3 d of infusion. This schedule provided 12 representative samples of ruminal, duodenal, and fecal contents taken at 2-h intervals.

Ruminal samples were immediately squeezed through two layers of cheesecloth. The filtrate was subsampled by taking 2 ml for protozoa counts and 20 ml for viscosity determinations. Ruminal particles obtained by squeezing were blended (400 g of particles plus 400 ml of 0.9% NaCl) in a Waring blender (Waring Products Division, New Hartford, CT) for 1 min and then squeezed through four layers of cheesecloth. Both filtrates from squeezed and strained homogenate were mixed, centrifuged ($800 \times g$ for 10 min at 4°C) to remove protozoa and feed particles, and the supernatant was centrifuged ($27,000 \times g$ for 30 min at 4°C) to obtain a mixed ruminal bacteria pellet. Bacterial isolates were accumulated by period, freeze-dried, and ground with a mortar and pestle. Subsamples of the ground bacterial composites were further ground using a ball mill (Wig-L-Bug; Crescent Dental Mfg. Co., Lyons, IL) to a fine powder for determination of N content and ^{15}N enrichment.

Duodenal samples were pooled by cow for each period with a mixer (model MX-9100, Toshiba, Tokyo, Japan) and freeze-dried for chemical analysis. Fecal samples were also pooled by cow for each period. The pH in feces was measured directly with a pH meter, then fecal samples were dried at 55°C and ground through a 1-mm screen for chemical analyses.

In Situ Measurements

The ruminal digestive kinetics of processed barley grain and barley silage were determined in sacco. Barley silage samples were collected fresh each period and chopped in 750-ml increments for 1 min in a household food processor. Five grams of processed barley (as fed) or 15 g of prepared barley silage were weighed into small bags (10×20 cm) made of monofilament PeCAP polyester (pore size, $51 \pm 2 \mu\text{m}$; B. & S.H. Thompson, Ville Mont-Royal, QC, Canada). Bags were heat-sealed and placed in large (20×30 cm) mesh retaining sacs with 3×5 -mm pores that permitted ruminal fluid to percolate freely. Duplicate nylon bags were placed in the rumen for 0, 2, 4, 6, 12, 24, and 48 h for grain or 0, 6, 12, 24, 48, 72, and 120 h for barley silage. Upon removal, bags were washed under running tap water until the effluent was clear and then were dried at 55°C for 48 h. Bags and contents were weighed and residues were ground to pass a 1-mm screen (Intermediate Mill, Arthur Thomas Co., Philadelphia, PA) and stored for analyses.

Kinetics of DM disappearance in situ was estimated by the nonlinear regression procedure of SAS (29). For each cow and type of feed, the following model was fitted to the percentage of DM disappearance:

$$y = a + b(1 - e^{-(t-L)}) \text{ for } t > L,$$

where

- a = soluble fraction (percentage)
- b = slowly digestible fraction (percentage)
- c = fractional rate of disappearance (per hour)
- L = lag time (hours) and
- t = time of incubation (hours).

Effective ruminal degradability (**ERD**) of DM was calculated by the equation

$$a + b \times c/(c + k)$$

where k = fractional passage rate (0.028/h was the average measured in the present study).

Chemical Analyses

Feed DM was determined by oven drying at 55°C for 48 h. Analytical DM content of the samples was determined by drying at 135°C for 3 h (2). The OM content was calculated as the difference between DM and ash contents. The NDF and ADF contents were determined by the methods described by Van Soest et al. (33) with amylase and sodium sulfite used in the NDF procedure. Starch was determined by enzymatic hydrolysis of α -linked glucose polymers as described previously (7). Contents of Cr, Yb, and Co in the samples were determined by atomic absorption spectrophotometry according to the AOAC (2). Ruminal VFA were separated and quantified by gas chromatography (Varian 3700; Varian Specialties Ltd., Brockville, ON) with a 15-m (0.53-mm i.d.) fused silica column (DB-FFAP column; J and W Scientific, Folsom, CA). Ammonia content of ruminal samples was determined by the method described by Weatherburn (34) modified to use a plate reader. Ruminal and duodenal viscosity was measured as follows: the samples were thawed and centrifuged at $1000 \times g$ for 10 min. The supernatant was transferred into a vial and kept in a water bath at 39°C until measurements were made. The viscosity of the supernatant was determined at 39°C with a tube-type falling ball viscometer (Gilmont Instruments, Barnant Company, Barrington, IL). The viscosity was calculated using the equation:

$$\mu = K(pt - p)^*t$$

where

μ = viscosity in centipoises (cp)
 pt = density of ball (g/ml)
 p = density of liquid
 t = time of descent (min) and
 K = viscosity constant.

Ruminal protozoa were counted with the aid of a microscope after mixing ruminal fluid with methyl green-formalin-saline solution. Content of CP in the samples and enrichment of ^{15}N in the bacteria isolated from the rumen and in duodenal digesta was determined by flash combustion (Carlo Erba Instruments, Milan, Italy) with isotope ratio mass spectrometry (VG Isotech, Middlewich, England). The volume weight of barley grain, on a DM basis, was determined by measuring the weight (kg) of 0.5 L (Seedburo, Chicago, IL) multiplied by the proportion of DM in the sample, and then using a conversion factor of 200 to obtain kg of DM/hl.

Calculations and Statistical Analyses

Flows of DM to the duodenum and DM excreted in feces were calculated by dividing Yb-infused (grams of Yb per day) by Yb concentration (grams of Yb per kilogram of DM) in the duodenal digesta or feces, respectively. Flows of other nutrients to the duodenum or feces were calculated by multiplying DM flow by their concentration in duodenal or fecal DM. Ruminal microbial protein synthesis for each cow was estimated by the ratio of ^{15}N flow at the duodenum to ^{15}N concentration of mixed ruminal bacteria. Kinetics of liquid and forage passage out of the rumen were estimated by the nonlinear regression procedure of SAS (29) for each cow and period from Co and Cr concentrations in ruminal samples, respectively.

For each period, means for individual cows were calculated for all variables, including feed intake, milk production, nutrient digestibility, rate of passage, and ruminal fermentation. Data were analyzed with the general linear models procedure of SAS (29) to account for effects of cow, period, and diet. All variables were tested for linear and quadratic effects in relation to extent of barley processing.

RESULTS

Characteristics of Barley Grain

Dry matter content of processed barley grain was reduced by about 4 percentage units due to steaming and subsequent drying compared with original whole grain (Table 2). However, processing had no major impact on nutrient content of processed barley. As barley was rolled more flatly, its volume weight decreased. Standardized barley has a volume weight of 51 kg/hl but typically can range from 44 to 57 kg/hl (DM basis). Therefore, volume weight of processed barley will be affected by the initial volume weight of the whole grain. Furthermore, words like "coarse" and "flat" are too vague to define a particular degree of processing. The use of PI to describe processing overcomes these limitations. The PI decreases as grain is processed more extensively. The PI for coarse (81%) and flat barley (55.5%) used in this experiment represented the extremes attainable by steam-rolling. We have observed in western Canada that the PI of commercially steam-rolled barley fed to dairy cattle typically ranges between 60 and 80%.

Kernel thickness was decreased and kernel width was increased with decreasing PI of barley. Consequently, the proportion of DM retained on the top screen was increased with decreasing PI of barley, indicating that these kernels were flatter than those of coarsely

rolled barley. However, the proportion of DM retained by the two top screens (3.35 and 2.36 mm) was 98.5, 98.6, 98.4, and 96.7% for coarse, medium, medium-flat, and flat barley, respectively. Thus, increased processing shifted the proportion of kernels retained on the 2.36-mm screen to the 3.35-mm screen. The small particle fraction that passed through the 2.36-mm screen was minimal and varied from 1.6 to 3.4% with decreasing PI of barley, indicating that increasing the extent of processing did not substantially alter the proportion of fines. Geometric mean length was varied similarly to that of the proportion of DM retained on the 3.35-mm screen since it was dependent on screen size used and DM retained on the screen. Because rolling grain does not decrease kernel particle size, sieving techniques do not differentiate the range in extent of processing that can be determined visually or by using PI.

Intakes and Digestibility of DM, OM, Starch, NDF, and ADF

Dry matter intake, either expressed as kilograms per day or percentage of BW, was quadratically increased while the apparent digestibility of DM in the total tract was linearly increased with decreasing PI of barley (Table 3).

Because DMI increased quadratically, intakes of OM and starch were also quadratically increased with decreasing PI of barley. However, ruminal digestibility of OM was not affected by PI of barley. Thus, amount of OM truly fermented in the rumen was quadratically increased as a result of increased intake, rather than a change in ruminal digestibility. Postruminal digestibility of OM, or OM digested in the intestine expressed as percentage of OM arriving at the duodenum, was linearly increased with increased grain processing. Similarly, digestibility of OM in the total tract was linearly increased. However, there was no further increase in digestibility of OM either in the intestine or in the total tract when medium-flat barley was replaced by flatly rolled barley in the diets.

Starch digestion in the rumen was not statistically affected by grain processing, although ruminal digestion was numerically lower for cows fed coarsely rolled barley (61.0%) than for cows fed the other barleys, for which ruminal starch digestibilities were similar (70.5%). Postruminal starch digestibility, expressed as a percentage of starch passing to the duodenum, was linearly increased from 39.2% for cows fed a diet containing coarsely rolled barley to 78.8% for cows fed a diet containing medium-flat rolled barley. Consequently, digestibility of starch in the total tract was linearly increased as PI of barley fed to cows was reduced. However, there was no further increase of total

tract starch digestibility when medium-flat barley was replaced by flatly rolled barley in the diet.

Similar to DMI, intakes of NDF and ADF were quadratically increased when PI of barley was decreased (Table 4). Although digestion of NDF and ADF in the rumen was not affected by grain processing, the amounts of NDF and ADF digested in the rumen followed a quadratic pattern similar to that of intakes of NDF and ADF. There was more fiber digested in the rumen for cows fed medium or medium-flat rolled barley than for cows fed coarsely or flatly rolled barley. However, opposite results were observed for digestibility of NDF in the intestine ($P < 0.12$). Consequently, the digestibilities of NDF and ADF in the total tract were not affected by degree of barley processing.

N Metabolism

As observed for other nutrients, intake of N was quadratically increased when PI of barley in the diets was decreased (Table 5). Passage of total N and NAN to the duodenum tended ($P < 0.11$) to follow a quadratic pattern similar to that of intake of N. This quadratic pattern was the result of nonsignificant increases in the flow of both RUP ($P < 0.19$) and microbial N ($P < 0.17$).

Digestion of N in the rumen was not affected by PI of barley. However, digestibility of N in the intestine, expressed as percentage of N flow to duodenum, was linearly increased with increased processing. Consequently, digestibility of CP in the total tract was linearly increased ($P < 0.06$) by reducing PI of barley in the diet.

Rumen Fermentation

The pattern of diurnal fluctuation of ruminal pH was similar among the treatments (Figure 1). The highest pH values were observed just before the morning feeding, and the lowest pH values were between 2100 and 0300 h. Although differences in ruminal pH among the treatments were not always statistically significant throughout the 24-h period, the area under pH curve was linearly decreased, and the hours during which rumen pH were below 6.2 or 5.8 were linearly increased with decreasing PI of barley (Table 6). Cows fed medium-flat or flatly rolled barley had much longer time that ruminal pH was below 5.8 than cows fed coarse or medium rolled barley.

On average, mean ruminal pH was decreased linearly with decreasing PI of barley, and the decrease was largest when coarsely rolled barley was replaced by medium rolled barley (Table 6). The linear decrease in ruminal pH corresponded to a linear increase ($P < 0.11$)

Table 3. Intake and digestibility of DM, OM, and starch in the digestive tract of dairy cows fed diets containing processed barley.

Item	PI ¹ (%)	Diet				SE	Contrasts, <i>P</i> <	
		Coarse	Medium	Medium -flat	Flat			
		81	72.5	64	55.5		Linear	Quadratic
DMI								
kg/d		18.7	21.4	21.7	20.1	0.6	0.12	0.01
% of BW		2.72	3.14	3.16	2.91	0.1	0.22	0.02
BW, kg		688	684	688	697	6	0.30	0.34
BCS		2.7	2.8	2.8	2.7	0.05	0.72	0.22
ADTT, ² %		62.4	63.9	70.3	69.8	1.9	0.02	0.62
OM								
Intake, kg/d		17.1	19.7	19.9	18.5	0.6	0.12	0.01
RFOF, ³ kg/d		9.7	12.0	12.2	11.1	0.6	0.17	0.04
Flow to duodenum, kg/d								
Total		9.7	10.9	11.0	10.5	0.7	0.49	0.28
Microbial		2.30	3.28	3.26	3.1	0.39	0.21	0.19
Digestibility, %								
Ruminal								
Apparently		42.7	44.5	44.0	43.7	3.2	0.88	0.75
Truly		56.2	60.9	60.7	60.4	2.9	0.37	0.41
Postruminal								
% of OM intake		20.6	20.9	28.0	28.0	3.6	0.12	0.97
% of flow to duodenum		35.3	37.9	49.7	49.8	4.5	0.03	0.79
ADTT, %		63.4	65.5	72.0	71.7	1.9	0.01	0.54
Starch								
Intake, kg/d		6.2	7.1	7.2	6.7	0.2	0.06	0.01
Flow to duodenum, kg/d		2.37	2.09	2.08	1.99	0.28	0.39	0.74
Digestibility, %								
Ruminal		61.0	70.3	70.5	70.7	4.6	0.21	0.37
Postruminal								
% of starch intake		17.0	13.8	23.1	22.2	4.6	0.27	0.81
% of flow to duodenum		39.2	47.5	78.8	75.9	5.9	0.01	0.38
ADTT, %		78.0	84.1	93.6	92.9	1.7	0.01	0.1

¹Processing index is the volume weight of the barley after processing, expressed as a percentage of its volume weight before processing.

²Apparently digested in the total tract.

³Organic matter truly fermented in the rumen.

in total ruminal VFA concentration with reducing PI of barley in the diets fed to cows. Feeding medium-rolled barley significantly reduced the molar percentage of acetate but increased the molar percentage of propionate, which resulted in a large decrease (0.71 units) in the acetate:propionate ratio, compared to feeding coarsely or medium rolled barley.

The concentration of NH₃ N in ruminal fluid increased when coarse rolled barley was replaced by medium rolled barley in the diets fed to cows. However, NH₃ N dramatically declined when medium rolled barley was replaced by medium-flat or flatly rolled barley. The viscosity of the ruminal fluid was not different among the treatments, but a quadratic pattern was detected for the viscosity of duodenal contents with decreasing PI of barley.

Chewing Time and Rate of Passage

Daily total eating time was similar for cows fed diets differing in PI of barley (Table 7). However, eating time,

expressed as minutes per unit of DM ($P < 0.12$) or NDF ($P < 0.07$), tended to be shorter for cows fed medium or medium-flat barley than for cows fed coarsely or flatly rolled barley. Similar results ($P < 0.01$) were also observed for ruminating time when expressed as minutes per unit of DM or NDF. In addition, daily total ruminating time linearly increased as PI of barley decreased, reflecting the larger particle size of flatter barley.

Liquid outflow rate from the rumen was not affected by PI of barley (Table 7). However, particle outflow rate from the rumen was linearly ($P < 0.07$) increased as PI of barley in the diets fed to cows was reduced.

Production and Composition of Milk

Milk production (actual, 4% FCM or SCM) was affected by PI of barley in the diets fed to cows (Table 8). Compared to cows fed coarsely rolled barley, milk yield was increased by 9.8, 20.3, and 13.3% for cows fed medium, medium-flat, and flatly rolled barley, respec-

tively. Hence, milk production increased when medium or flatly rolled barley was replaced by medium-flat barley.

Impact of PI of barley on milk fat and lactose contents was negligible; however, milk protein content was highest for cows fed flatly rolled barley and lowest for cows fed coarsely rolled barley.

In contrast to the milk production, milk efficiency was quadratically affected ($P < 0.07$) as PI of barley was decreased. Cows fed flatly rolled barley had the best milk efficiency, while the lowest milk efficiency was observed for cows fed medium rolled barley.

In Situ Ruminal Digestion Kinetics

For processed barley grain, the soluble fraction increased linearly, but the potential degradable fraction was linearly reduced ($P < 0.07$) when PI of barley was decreased (Table 9). Ruminal degradation rate and ERD also increased linearly with reduction of PI of barley.

For barley silage, impact of PI of barley in the diets on the soluble fraction was small, even though these were statistically significant. The potential degradable

fraction tended to ($P < 0.14$) be quadratically affected by extent of barley processing, while ruminal degradation rate was quadratically decreased. Reducing PI of barley in the diets fed to cows quadratically decreased ERD of barley silage in situ.

DISCUSSION

Whole barley grain is not efficiently digested by cattle; consequently, barley is usually processed prior to feeding to breach the hull and pericarp, thereby promoting access of ruminal microorganisms to the endosperm (19). Steam-rolling is often preferred over dry-rolling because it is easier to control the resulting kernel thickness and minimize the amount of fines. This study examined the extent to which rolling barley grain, as measured by PI, can be used to regulate its rate and extent of digestion. While it is generally recognized within the feed industry that extent of processing can be used to regulate rate of digestion of barley grain in the rumen, this is the first study to our knowledge to quantitatively examine this concept for dairy cows.

Replacing coarsely rolled barley with medium-flat barley in the diet of dairy cows increased milk produc-

Table 4. Intake and digestibility of NDF and ADF in the digestive tract of dairy cows fed diets containing processed barley.

Item	PI ¹ (%)	Diet ¹				SE	Contrasts, <i>P</i> <	
		Coarse	Medium	Medium -flat	Flat			
		81	72.5	64	55.5		Linear	Quadratic
NDF								
Intake, kg/d		7.4	8.2	8.2	7.5	0.4	0.85	0.07
Flow to duodenum, kg/d		3.92	4.16	3.80	3.97	0.29	0.87	0.91
Flow to feces, kg/d		3.23	3.84	3.39	3.26	0.26	0.76	0.20
Digestibility								
Ruminal								
kg/d		3.43	4.08	4.41	3.50	0.22	0.60	0.02
%		47.7	48.0	53.2	47.1	2.9	0.50	0.31
Postruminal								
% of NDF intake		8.9	5.2	5.3	9.2	2.1	0.93	0.12
% of flow to duodenum		15.2	7.6	9.7	17.0	3.8	0.68	0.10
ADTT ² , %		56.6	53.2	58.4	56.3	2.9	0.75	0.85
ADF								
Intake, kg/d		4.1	4.7	4.5	4.1	0.2	0.67	0.03
Flow to duodenum, kg/d		2.67	2.79	2.49	2.67	0.21	0.76	0.89
Flow to feces, kg/d		2.24	2.73	2.46	2.40	0.17	0.80	0.16
Digestibility, %								
Ruminal								
kg/d		1.47	1.89	2.03	1.40	0.18	0.96	0.03
%		36.6	38.7	44.1	34.9	4.4	0.99	0.24
Postruminal								
% of NDF intake		9.7	2.5	1.1	5.7	3.9	0.48	0.19
% of flow to duodenum		10.2	1.8	0.4	6.9	5.0	0.63	0.19
ADTT ² , %		46.3	41.2	45.2	40.6	3.8	0.47	0.96

¹Processing index is the volume weight of the barley after processing, expressed as a percentage of its volume weight before processing.

²Apparently digested in the total tract.

Table 5. Intake of N, and flow and digestibility of nitrogenous compounds of the digestive tract of dairy cows fed diets containing processed barley.

Item	PI ¹ (%)	Diet				SE	Contrasts, <i>P</i> <	
		Coarse	Medium	Medium -flat	Flat			
		81	72.5	64	55.5		Linear	Quadratic
N intake, g/d		504.5	573.0	565.7	549.4	13.4	0.08	0.02
Flow to duodenum								
Total N								
g/d		477.9	610.0	588.0	543.7	47.0	0.44	0.11
% of N intake		94.2	105.6	105.7	97.5	7.1	0.76	0.22
NAN								
g/d		459.0	591.7	573.4	528.2	46.6	0.40	0.11
% of N intake		90.5	102.4	103.1	94.8	7.1	0.69	0.20
Feed + endogenous N								
g/d		252.4	304.7	278.8	253.3	26.3	0.85	0.19
% of NAN		54.9	52.0	48.6	47.6	3.3	0.14	0.78
% of N intake		49.6	53.2	49.2	44.8	4.2	0.36	0.37
Microbial N								
g/d		206.7	287.0	294.5	275.0	32.2	0.19	0.17
% of NAN		45.1	48.0	51.4	52.4	3.3	0.14	0.78
% of N intake		40.9	49.2	53.8	50.0	5.6	0.25	0.32
Microbial N efficiency, g/kg OMTD ²		21.7	23.5	25.2	24.6	2.5	0.38	0.64
Digestibility								
Ruminal, %								
Apparently		9.5	-2.5	-3.1	5.2	7.1	0.69	0.20
Truly		50.4	46.8	50.8	55.2	4.2	0.36	0.37
Postruminal								
% of N intake		60.5	72.1	75.8	69.8	5.7	0.26	0.17
% of N flow to duodenum		62.6	67.2	71.2	70.4	1.8	0.02	0.19
ADTT ³ , %		66.3	66.4	70.2	72.3	2.1	0.06	0.67

¹Processing index is the volume weight of the barley after processing, expressed as a percentage of its volume weight before processing.

²Organic matter truly digested in the rumen.

³Apparently digested in the total tract.

tion, which we attribute to the increase in both DMI and nutrient digestibility in the total tract. In contrast, the higher milk yield of cows fed medium-flat rolled

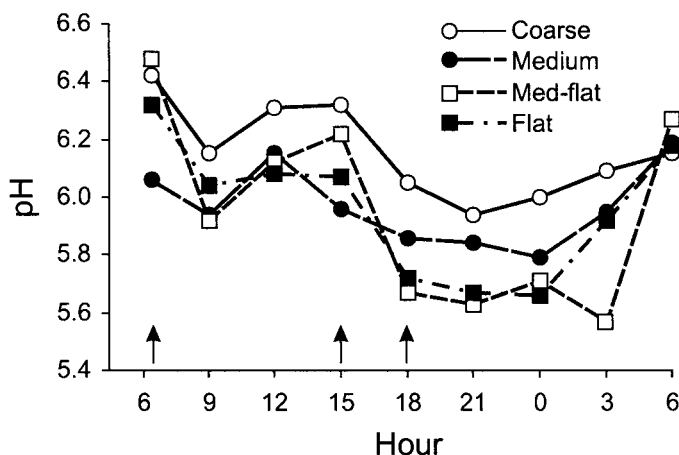


Figure 1. Diurnal fluctuation of ruminal pH in lactating dairy cows fed diets containing processed barley: coarse (○), medium (●), medium-flat (□), and flat (■). Each point represents the mean of four observations (SEM = 0.04) and the arrow designates feeding time.

barley compared with cows fed medium rolled barley resulted only from increased digestibility. Furthermore, the lower milk yield of cows fed flatly rolled barley compared to cows fed medium-flat rolled barley was mainly due to the decline in DMI because nutrient digestibilities were similar for both these groups of cows.

The importance of appropriately processing barley grain for use in dairy cow diets has been observed by others (16, 32, 36). Laksesvela (16) examined the significance of fines resulting from grinding a mixture of barley and oat grain and concluded that a medium grind was slightly superior for milk production than a coarse grind. Valentine and Wickes (32) reported that cows fed whole barley grain produced less milk than did cows fed rolled barley, because whole barley was less digestible than processed barley (13). Yang et al. (36) reported that cows fed hull-less barley (hull-less barley varieties have a seed coat that is loosely attached and easily removed during harvesting) with a PI of 82% produced less milk than did cows fed conventional barley with a PI of 68%. In contrast, when the PI of hull-less barley was reduced to 73%, cows fed hull-less barley produced more milk than did cows fed barley. In that study, in-

Table 6. Characteristics of ruminal fermentation in dairy cows fed diets containing processed barley.

Item	PI ¹ (%)	Diet				SE	Contrasts, <i>P</i> <	
		Coarse	Medium	Medium -flat	Flat			
		81	72.5	64	55.5		Linear	Quadratic
pH								
Average		6.16	5.97	5.95	5.96	0.03	0.02	0.06
Area under curve (pH × h/d)		147.4	142.9	141.7	142.3	1.0	0.01	0.04
<6.20, h		12.9	20.4	18.1	18.3	1.5	0.08	0.05
<5.80, h		3.7	6.1	9.0	10.6	1.4	0.01	0.77
Lowest		5.74	5.46	5.31	5.43	0.10	0.05	0.10
VFA								
Total, mM		125.3	136.3	137.8	137.8	4.6	0.11	0.27
mol/100 mol								
Acetate (A)		65.7	65.9	61.6	64.3	0.7	0.04	0.14
Propionate (P)		20.6	19.6	24.5	22.1	0.9	0.05	0.45
Butyrate		9.6	10.3	10.3	9.9	0.2	0.47	0.06
A:P		3.32	3.38	2.64	2.99	0.15	0.04	0.37
NH ₃ N, mg/L		140.0	160.7	120.0	116.4	8.1	0.02	0.18
Protozoa count (×10 ⁶)		1.75	2.09	1.79	1.86	0.15	0.97	0.40
Viscosity, cp								
Ruminal		1.65	1.63	1.65	1.64	0.04	1.00	0.92
Duodenal		1.33	1.40	1.36	1.30	0.03	0.38	0.06
Feces								
pH		7.32	7.26	7.24	7.26	0.07	0.53	0.57
Whole grain, % of DM		31.5	21.0	10.3	10.0	1.3	0.01	0.01

¹Processing index is the volume weight of the barley after processing, expressed as a percentage of its volume weight before processing.

creased milk production due to reduced PI of hull-less barley resulted from improved digestibility rather than increased DMI.

Improving animal performance by manipulating PI of barley has also been reported for beef cattle. Hironaka et al. (14) reported that steam-rolled barley rolled to a medium kernel thickness (PI = 82%) resulted in better average daily gain than when thin (PI = 74%), coarse (PI = 92%), or whole (PI = 100%) barley was fed.

From that study, it appears that the optimum PI of barley for feedlot cattle is higher than the optimum PI observed in the current study for dairy cows. Similarly, Owens et al. (26) reported for feedlot cattle that barley of a medium flake thickness was superior to either a barley of thicker or thinner flake for average daily gain, DMI, and feed efficiency. The reason that optimum PI is higher for feedlot cattle fed finishing diets than for dairy cows is thought to be due to the much lower con-

Table 7. Characteristics of chewing activity in dairy cows fed diets containing processed barley.

Item	PI ¹ (%)	Diet				SE	Contrasts, <i>P</i> <	
		Coarse	Medium	Medium -flat	Flat			
		81	72.5	64	55.5		Linear	Quadratic
Eating								
Min/d		255	265	265	266	15	0.62	0.77
Min/kg DM		13.9	12.6	12.4	13.3	0.6	0.48	0.12
Min/kg NDF		35.1	33.1	32.5	35.6	1.1	0.85	0.07
Ruminating								
Min/d		443	421	468	495	15	0.03	0.16
Min/kg DM		23.9	19.8	21.7	25.0	0.8	0.22	0.01
Min/kg NDF		60.7	51.9	56.9	66.9	2.5	0.08	0.01
Eating/ruminating		0.58	0.64	0.59	0.54	0.03	0.21	0.14
Ruminal outflow rate, %								
Liquid		12.1	12.5	12.9	15.7	2.1	0.27	0.58
Particles		2.22	3.32	3.40	3.18	0.29	0.07	0.07

¹Processing index is the volume weight of the barley after processing, expressed as a percentage of its volume weight before processing.

Table 8. Milk production and composition of lactating dairy cows fed diets containing processed barley.

Item	PI ¹ (%)	Diet				SE	Contrasts, <i>P</i> <	
		Coarse	Medium	Medium -flat	Flat			
		81	72.5	64	55.5		Linear	Quadratic
Milk production, kg/d								
Actual		25.6	28.1	30.8	29.0	0.4	0.01	0.01
4% FCM		25.2	27.4	29.5	28.4	0.3	0.01	0.01
SCM		24.4	26.9	29.1	28.0	0.3	0.01	0.01
Milk composition, %								
Fat		3.93	3.89	3.78	3.90	0.06	0.5	0.25
Protein		3.15	3.30	3.29	3.34	0.02	0.01	0.05
Lactose		4.56	4.56	4.58	4.57	0.01	0.12	0.48
Milk efficiency								
Milk/DMI		1.38	1.31	1.41	1.46	0.04	0.12	0.07
FCM/DMI		1.37	1.28	1.36	1.45	0.04	0.07	0.13

¹Processing index is the volume weight of the barley after processing, expressed as a percentage of its volume weight before processing.

centration of forage (i.e., usually < 10% forage, DM basis), or effective fiber, in diets fed to feedlot cattle. It is possible that less extensively processed barley would be desirable in dairy diets formulated to supply less than adequate effective fiber (8).

The lack of effect of barley PI on milk fat percentage was probably due to consistent ruminal starch and fiber digestion among treatments. The linear increase in milk protein content with decreased PI of barley is attributable to more available protein due to higher protein intake and digestibility. Burgess and Nicholson (10) reported that increasing the level of dietary CP from deficient to adequate levels (10 vs. 13 and 16%

CP) increased milk protein content. In addition, the larger amount of starch digested in the rumen with decreased PI of barley provided more propionate for glucose synthesis, which may have spared protein and increased milk protein content.

Of the three major components of milk, lactose tends to be the least amenable to manipulation by dietary means. This is confirmed by the present and previous results for diets comprised of barley grain (35).

The trend towards improving feed efficiency by using more extensively processed barley is in agreement with others (13, 18). This improvement was due to the increase in nutrient digestibility. Lower DMI of cows fed

Table 9. In situ ruminal DM digestion kinetics of barley grain and barley silage

Parameters ²	PI ¹ (%)	Diet ¹				SE	Contrasts, <i>P</i> <	
		Coarse	Medium	Medium -flat	Flat			
		81	72.5	64	55.5		Linear	Quadratic
Grain								
a, %		1.3	1.5	2.8	6.6	0.9	0.01	0.09
b, %		91.0	92.4	87.4	78.2	4.3	0.07	0.26
c, %		2.41	3.76	5.30	7.61	0.46	0.01	0.34
Lag time, h		3.84	1.26	0.01	0.09	0.26	0.01	0.01
ERD, %		42.3	53.8	59.4	63.6	0.8	0.01	0.01
Barley silage								
a, %		32.3	33.5	33.0	33.3	0.1	0.01	0.03
b, %		47.3	45.9	49.3	44.1	1.1	0.25	0.14
c, %		2.98	2.41	2.04	2.57	0.15	0.06	0.01
Lag time, h		0.54	0	0	0	0.27	0.23	0.36
ERD, %		56.6	54.7	53.5	54.5	0.6	0.03	0.06

¹Processing index is the volume weight of the barley after processing, expressed as a percentage of its volume weight before processing.

²Parameters were calculated from the fitted equation $P = a + b(1 - e^{-c(t-L)})$ for $t > L$, where P = percentage of DM disappearance from the bag at time t , a = soluble fraction, b = slowly degradable fraction, c = fraction rate constant at which b is degraded, L = lag time (hours), and t = time of incubation (hours). Effective ruminal degradability (ERD) was calculated using equation $a + bc/(c + k)$, where $k = 0.028/h$.

coarsely rolled barley compared with those fed more extensively processed barley was likely due to the combined effects of slightly lower ruminal digestibility and lower rate of particulate passage out of the rumen. We did not expect lower intake for cows fed coarsely rolled barley and this result is opposite to other reports (13, 18), in which feed intake was observed to increase when grain of coarser particle size was fed. However, in those experiments, beef cattle were used and the diets contained 85 to 97% barley grain. In contrast, the forage represented about 47% of the dietary DM in the present study. In addition, lower particulate rate of passage for the diet containing coarsely rolled barley was not expected because we observed previously that a diet containing coarsely rolled hull-less barley passed out of the rumen 6% faster than medium rolled barley (35).

Lower DMI of cows fed flatly rolled barley compared with medium-flat barley supports the work of others (13, 18, 26). This effect might be attributed to the increased acidity in the rumen of cattle offered finely rolled barley. Hironaka et al. (13) observed more feedlot cattle off-feed due to digestive disturbances among those fed thinly rolled barley or medium rolled barley than among those fed coarsely rolled barley or whole barley. However, our data do not support this mode of action, as pH variables were similar for cows fed medium-flat barley (highest intake) and those fed finely processed barley. If indeed the decrease in intake resulting from extensive processing of barley is due to increased ruminal fermentation and a subsequent decrease in fiber digestion, the effect must be manifested a manner that is more subtle than is apparent from standard methods used to measure ruminal acidity and digestion.

The estimates of OM digestibility within the rumen are in good agreement with our previous study (35), but slightly higher than those of others (25) for dairy cows fed similar diets. Decreasing the PI did not increase ruminal digestibility of OM, probably because DMI increased, which resulted in a quadratic increase in the amount of OM digested in the rumen. In a study that examined the influence of extent of barley processing on site and extent of digestion for feedlot cattle, barley was steam-rolled to a PI of 30 or 59% and the diets contained 10% forage (DM basis) (37). That study also showed that there was no difference in OM or starch digestion in the rumen, albeit the barley was highly processed (PI < 60) relative to other studies (PI > 70). However, in the study of Zinn (37), DMI was not affected by degree of barley processing. In the current study, lower digestibility of starch in the rumen for cows fed coarsely rolled barley is similar to our previous results (35), in which hull-less barley was coarsely

rolled (PI = 81.9%) and its ruminal starch digestibility was only 46.7%.

The digestion of OM or starch in the intestine was affected by barley processing. The intestinal digestibilities of OM or starch were lowest for cows fed coarsely rolled barley and highest for cows fed medium-flat or flatly rolled barley. Intestinal digestibility of OM and starch was also relatively low for cows fed medium rolled barley, indicating that rumination of barley grain was not sufficient to compensate for ineffective processing (6) and intact barley grains pass quickly through the intestine. Yang et al. (35) observed that lower ruminal digestibility was compensated for by higher intestinal digestion when cows were fed corn, but not for cows fed coarsely rolled hull-less barley. In that study, authors concluded that digestion of barley in the intestine is positively related to its ruminal digestion: increasing particle size of grain compromises its intestinal digestion. In the present study ruminal digestibility was similar for all barleys, but the digestion of OM or starch in the intestine and in the total tract was directly proportional to extent of barley processing from coarse to medium-flat processing. Further processing did not improve starch digestion because the starch contributed by silage was poorly digested in the rumen and at least 10% of the fecal DM was in the form of whole or slightly damaged grain. The kernels after ingestive mastication recovered in feces were assumed to originate from the silage because the grain portion of the diet was processed in a manner that resulted in few intact kernels after ingestive mastication, at least for the diets containing medium-flat or flatly rolled barley grain.

Differences in OM digestibility in the total tract among treatments can be attributed mainly to differences in starch digestibility, as fiber digestibility in the rumen or in the total tract was not affected by PI of barley. Similarly, Morgan et al. (22) reported that NDF digestibility in the rumen of cows was not affected by barley processing. Activity of the predominant fibrolytic ruminal bacteria such as *Fibrobacter* spp. and *Ruminococcus albus* declines rapidly when pH falls below 6.0 in vitro (28). If a parallel situation occurs in vivo, it should be surprising to observe fiber digestion at all under the ruminal conditions observed in this study. On the contrary, ruminal fiber digestion was unaffected by the relatively low ruminal pH when extensively processed barley was fed. This indicates that other fibrolytic organisms must be contributing significantly to ruminal fiber digestion. Forster et al. (11) found that over 60% of rumen bacterial species have not been fully characterized and some of these species may be fibrolytic. The role of what has been considered to be minor

populations of rumen bacteria on fiber digestion needs to be revisited.

The benefits of shifting site of starch digestion from the rumen to the intestine, or vice versa, is unclear. McAllister et al. (20) suggested that increasing the post-ruminal supply of starch to the small intestine may improve feed efficiency by reducing the loss of energy through methane or heat. Owens et al. (27) reported that starch digested in the rumen is used only 70% as efficiently as starch digested in the small intestine. In contrast, Taniguchi et al. (30) found that when starch was digested in the rumen it provided 6 MJ more energy daily to steers than if the starch was digested in the small intestine. The advantages and disadvantages of ruminal digestion of starch were discussed by Hunt (15) and appear to be rather closely balanced. Furthermore, review of dairy cow studies did not suggest any clear relationship between milk production and site of starch digestion. Nevertheless, Owens et al. (27) identified numerous factors that affect the partition of starch digestion between the rumen and the small intestine, and it is not fully understood how this partitioning is influenced by feed processing.

It seems that there is a highly positive correlation between ruminal and intestinal digestion of starch for mechanical processed barley based on our previous (35) and present results. Increasing extent of barley processing favors both ruminal and intestinal digestion of barley grain and decreases the proportion of whole grain excreted in feces. Thus, shifting the site of barley digestion between the rumen and intestine by manipulating processing is not a viable option as it may be for corn.

It is generally accepted that efficiency of microbial protein synthesis is severely compromised when ruminal pH drops below 6.2 based on *in vitro* studies (28). However, microbial efficiency was not impaired due to pH in this study. In fact, efficiency of microbial protein synthesis for the medium-flat diet was numerically greater than the coarse diet, despite ruminal pH being below 6.2 more than 80% of the time. Clearly the relationship between ruminal pH and microbial protein synthesis needs to be refined to accommodate *in vivo* results obtained with lactating dairy cows.

In this study, cows on all diets chewed >11 h/d with ≥ 7 h/d of rumination. This result is consistent with other reports (5). High producing dairy cows consuming large quantities of DM tend to ruminate ≥ 6 h/d, unless a digestive upset occurs. The linear increase in rumination time with increased processing was likely due to the increase in fiber intake (4) combined with the fact that the kernels were larger. Expressing rumination time on the basis of DM and NDF intake revealed that diets containing either coarsely rolled barley or flatly

rolled barley were ruminated more than the other diets. In the case of the flatly rolled barley, increased rumination time per unit of feed may have resulted from the larger width and length of kernels. In the case of the coarsely rolled barley, increased rumination time per unit of feed was likely due to the increased need for particle size reduction of coarse barley kernels, as physical disruption of the hull due to processing was minimal. Extent of processing barley altered its effective fiber content, as assessed by chewing time. If the effective fiber value of medium rolled barley is assumed to be 34% (24), then the value for coarse barley would be 40%, medium-flat barley 37%, and that of flatly rolled barley 43%.

Although there is very limited information concerning the optimal degree of processing of barley grain, the present study and other reports (14, 18, 36) demonstrate that extent of barley processing has an influence on cattle performance. Failure to observe differences between coarse and thin-flaked barley fed to feedlot cattle by Zinn (37) was likely due to the extremely flat processing used. Optimum extent of rolling of barley for dairy cows fed diets with adequate effective fiber appears to correspond to a PI of 64%.

There is a need to develop suitable standardized techniques for measurement of extent of processing. Mathison et al. (18) listed methods currently used to assess extent of grain processing, including whole kernel count, sieving, kernel thickness, width and geometric particle size, and density. These measurements are useful in terms of describing the extent of barley processing, but they are not easily applied commercially. However, PI, as used in this study, seems to be a suitable method of measuring extent of barley processing due to its simplicity. Most grain processing facilities already have equipment for measuring the weight and volume of grains. This study demonstrates that PI measurement is a particularly reliable method to describe the extent of processing for steam-rolled barley and that PI is related to performance of dairy cows. The results of Mathison et al. (18) may discourage the use of PI to quantify the extent of grain processing; however, in that study the weight of processed barley was measured based on a concentrate mixture. Supplement may have filled the spaces between rolled barley kernels, thereby increasing volume weight of processed barley. This may explain the marginal difference in volume weight between slightly rolled and medium rolled barley in their study.

CONCLUSIONS

The degree of rolling of barley grain affected DMI, site and extent of nutrient digestibility, rumen fermenta-

tation, chewing activity, and milk production and composition of cows. Barley that was rolled to a medium-flat thickness produced the most milk, because of highest DMI and highest digestibility in the rumen and in the intestine. Feeding cows coarsely rolled barley was the least effective, because of low DMI combined with low digestibility in the total tract. In this study in which adequate effective fiber was offered, extensively rolled barley did not produce severe digestive disturbances, but DMI was compromised.

It is concluded that optimal rolling of barley for dairy cows results in a PI of 64%. Coarse rolling of barley, which is commonly done to reduce its ruminal rate of fermentation, is not recommended, as the physical barriers that limit ruminal digestion by ruminal microorganisms also limit enzymatic digestion in the small intestine. The PI, which is measured as the density of barley after processing as percentage of density before processing, is a reliable and practical method to quantitatively measure extent of processing.

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